14-Day Sprint Plan — Suicide Risk Detection Paper

Goal: convert the finished BUBT capstone thesis into a submittable conference/journal paper. **No new experiments, no new data collection** — the dataset (1,617 records), all 32 imputation×clustering pipelines, and the results are already done. This sprint is a writing, sourcing, and formatting job, scoped in SPRINT_TRACKER.md.

Start: **2026-08-17** · End: **2026-08-30** (14 calendar days). Adjust dates if you start on a different day — the day numbers matter more than the dates.

What already exists (verified this session)

Source	Location	Contains
Full thesis text	`prior-work/newly formated capstone report 2025.docx`	All 6 chapters, ~147k chars, **authoritative content** — intro, lit review, methodology, results, discussion, limitations, future work, and a **42-entry reference list** (plain text, end of doc)
LaTeX thesis skeleton	`prior-work/Capstone_Project_Report.zip`	BUBT thesis template: `main.tex`, `chapter1.tex`, `chapter2.tex`, `chapter3.tex` (chapters 4–6 were never converted to LaTeX — docx is ahead of the tex), `cover_page.tex`, `ref.bib` (only 2 placeholder entries — advisor's own papers, not the real 42), `image/methodology.png`, `image/method.png`
Notebooks + raw results	`prior-work/capstone-....zip` → `capstone/`	12 numbered notebooks (`0_preprocessing` → `11_comparing-clustering`), all imputed datasets (`.xlsx`), and `capstone/report/11_comparing-clustering.ipynb - Colab.pdf` (3.6 MB, PDF export of the final comparison notebook — likely already has publication-candidate figures rendered)
New literature (uncited by thesis)	`literature/*.pdf`	9 papers: 3 South Asia suicide-epidemiology reviews, 1 suicide-risk-detection perspective piece, 5 NLP/ML suicide/ideation-detection papers. None of these are in the thesis's own 42-ref list — they're a second, newer batch.

Source	Location	Contains
Author/team info	`cover_page.tex` / `Declaration.tex`	Title: *"A Machine Learning Approach to Identifying Influential Factors and Clustering Patterns in Suicide Occurrences"*. Authors: Mohammed Akash, Md Arifuzzan, Sajedul Islam, Komol Chandra Devnath, Anisul Haque Bhuiyan. Supervisor: Md. Mamun Hossain (BUBT CSE)
Best result	docx §6.1–6.2	Best pipeline = Agglomerative Clustering + MissForest (no-context), composite score 27.30/32, two-cluster solution that organically separated by data-collection cohort (2020 vs 2017–2025)
Full metrics table	docx Appendix, Table S1	All 32 pipelines' raw scores (balance, silhouette, Calinski-Harabasz, Dunn, Davies-Bouldin, Xie-Beni, composite) — ready to become a paper table

The `ref.bib` gap is real: the 42 sources exist only as plain-text citations at the end of the docx, not as BibTeX. Building the real `.bib` is on the critical path (Day 2).

Working structure for this sprint

Create a `paper/` folder inside `suicide-risk-detection/` (kept separate from `prior-work/`, which stays as the untouched thesis archive):

```
suicide-risk-detection/paper/
├── main.tex                venue template (chosen Day 1)
├── refs.bib                 real bibliography, built Day 2
├── sections/
│   ├── abstract.tex
│   ├── introduction.tex
│   ├── related-work.tex
│   ├── methodology.tex
│   ├── results.tex
│   ├── discussion.tex
│   └── conclusion.tex
└── figures/                  extracted/regenerated, publication-ready
```

Day-by-day

Day 1 — Scope, venue, skeleton

- Pick target venue/template and page limit. `ref.bib`'s two placeholder entries are the supervisor's own ICCIT/IJACSA papers — a strong hint the usual target is an IEEE double-column conference (e.g. ICCIT-style, ~6 pages). Confirm with the team/supervisor rather than assuming.
- Set up `paper/` with the chosen template's `main.tex` and empty `sections/*.tex` stubs.
- Write a bullet-point abstract skeleton (problem → gap → method → best result → contribution) to pin the paper's scope before drafting prose.
- Confirm author order and affiliations against `Declaration.tex`.

Day 2 — Bibliography

- Transcribe all 42 references from the docx's plain-text reference list into `refs.bib` as real BibTeX entries (replace the 2 placeholders).
- Cross-check numbering against in-text citations in the docx (`[1]`, `[2]`, `[12,13]`, etc.) so citation keys map 1:1 — the docx's own bracket numbers got mangled by the .docx → text extraction in a few spots, so verify each one against context, not just position.

Day 3 — Related work: fold in the new literature

- Read the 9 PDFs in `literature/`. Group them: (a) South Asia epidemiology — Arafat (Global Mental Health), Rahman et al. (BMJ Open protocol), Borah & Jose (F1000 protocol) — use these to strengthen the "burden of suicide in Bangladesh/South Asia" framing already in docx §2.2. (b) NLP/ML suicide-detection — Zhang et al. (transformer suicide notes), Fernandes et al. (NLP on clinical records), Ezerceli & Dehkharghani (deep nets on social media), Li et al. (visual behaviour pipeline), Ghanadian et al. (LLM synthetic data), Parsapoor et al. (perspective) — use these to sharpen the paper's novelty claim: *this study clusters structured, incomplete newspaper records rather than mining free text*, a genuinely different (and complementary) angle worth stating explicitly.
- Add these 9 as new `refs.bib` entries (now ~51 total).
- Draft `related-work.tex` — condense docx Chapter 2 (2.2–2.6) down from a thesis-length review to a paper-length one (roughly 500–800 words), positioned around the gap statement already in docx §2.6/§1.3.

Day 4 — Introduction

- Condense docx §1.1–1.4 (background, problem statement, objectives) into `introduction.tex` (~400–600 words). Keep the 6 research objectives as a tight paragraph, not a bulleted list, to fit paper conventions.
- State the paper's contributions explicitly (docx §1.6 already has these pre-written as 5 bullets — trim to 3–4).

Day 5 — Methodology

- Condense docx Chapter 3 into `methodology.tex`: dataset construction (760 → 1,617 records, 9 new attributes), cleaning + Geopy/Meteostat conditional filling, the 3 imputation methods × 2 predictor strategies (6 variants + 2 baselines = 8 datasets), the 4 clustering algorithms (32 pipelines total), and the composite balance-aware scoring metric.
- Reuse `image/methodology.png` and `image/method.png` from the tex archive if they're still accurate; otherwise redraw a compact pipeline diagram.

Day 6 — Extract figures & tables from notebooks

- Pull embedded output images directly from `capstone/11_comparing-clustering.ipynb` and/or `capstone/report/11_comparing-clustering.ipynb - Colab.pdf` — no need to re-run any notebook. Target: composite-score ranking chart, PCA cluster scatter for the winning pipeline (`agg_rf_no_context`), missingness before/after comparison.
- Turn the appendix Table S1 (already in the docx, all 32 rows) into a proper LaTeX table — full table as a supplementary table, a top-10 condensed version in the main results section.

Day 7 — Results

- Draft `results.tex` from docx Chapter 4/6.2: pipeline comparison summary, why MissForest > MICE/KNN, why imputed >> non-imputed baselines, why the balance metric mattered (the DBSCAN-noise-majority finding is a strong, concrete example — keep it), and the winning two-cluster solution.

Day 8 — Discussion

- Interpret the cluster-separation-by-cohort finding (docx §6.2 Finding 4) against the newly added related work: what does it imply given the NLP literature detects risk from *text*, while this study detects structure from *missingness patterns and metadata*? This contrast is the paper's most defensible novelty angle — make it explicit here, not just implied.

Day 9 — Limitations, ethics, future work

- Condense docx §5.5/§6.4 (6 limitations) to the 3–4 a reviewer will actually push on: data-source bias, no external validation, exclusion of Bangla free-text columns.
- Ethics: docx Chapter 5 already discusses societal impact/ethics for the thesis version — adapt for a paper-length ethics/data-statement paragraph (newspaper-sourced, publicly reported cases, no individually identifying claims made).
- Future work: trim docx §6.5's 6 directions to 2–3, prioritizing ones a reviewer would find most credible as immediate next steps (feature-driven cluster interpretation, Bangla NLP on the excluded free-text columns).

Day 10 — Abstract + conclusion + first full assembly

- Write final abstract (150–250 words, venue-dependent limit) from the Day 1 skeleton, now grounded in the actually-written sections.
- Write `conclusion.tex` from docx §6.6, cut to paper length.
- Assemble `main.tex` end to end; get a compiling draft with placeholder figure files if any extraction is still pending from Day 6.

Day 11 — Fit-to-venue pass

- Apply the chosen template precisely (columns, fonts, margins per Day-1 decision). Trim prose until the paper is under the page limit — this version is being cut down from a 147k-character thesis, so expect this pass to be substantial, not cosmetic.
- Finalize all figure/table captions and cross-references.

Day 12 — Internal review

- Circulate to co-authors and the supervisor (Md. Mamun Hossain) for a read-through — plan for this to take a day even if your own edits are done, since 5 co-authors is where most schedule slippage tends to happen.

- Specifically ask the supervisor to sanity-check the novelty framing against the new literature (Day 3) — that framing is new relative to the thesis and hasn't been vetted by them yet.

Day 13 — Self-plagiarism / duplicate-publication check

- Because this paper is substantially derived from an already-submitted university thesis, confirm the target venue's policy on prior-thesis material (most conferences allow it but expect a disclosure footnote, e.g. "based on the first author's undergraduate thesis at BUBT"). Add that disclosure if required. This is a real submission-blocking risk if skipped, not a formality.
- Full proofreading pass; verify every number in the paper (composite scores, missingness percentages, record counts) matches the docx/Table S1 source exactly — condensing 147k characters by hand is where transcription errors creep in.

Day 14 — Final polish + submission

- Final compile, reference check (all `refs.bib` entries actually cited, no orphaned citations), file-naming per venue's submission instructions.
- Cover letter (if required) and submission-portal upload.
- Keep this day as buffer — if Day 12's co-author review surfaces substantive change requests, they land here.

Biggest risks to this timeline

1. **Co-author/supervisor review latency (Day 12)** — 5 authors + 1 supervisor is the most likely place the 14 days slip. Send the draft for review as early as a stable draft exists, not literally on Day 12.
2. **Reference transcription (Day 2)** — 42 hand-typed citations from a docx, several missing their bracket numbers in the raw text, is fiddly and error-prone. Budget real time for it rather than treating it as mechanical.
3. **Venue page limit vs. thesis depth (Day 11)** — a 6-page IEEE format from a 147k-character, 6-chapter thesis is an aggressive cut. If the chosen venue allows 8–10 pages instead of 6, Day 11 gets much easier — worth factoring into the Day 1 venue decision.